Social Media Engagement Analysis Using Python

Project Overview
This project analyzes social media engagement data using Python. The objective is to clean, process, analyze, and visualize engagement metrics such as likes, comments, shares, impressions, watch time, and engagement rates. The project helps identify trends, audience behavior, and content performance through statistical analysis and data visualization.

Objectives
- Load and inspect social media engagement data.
- Clean missing and duplicate records.
- Create new features for better analysis.
- Perform exploratory data analysis (EDA).
- Calculate statistical measures.
- Visualize engagement patterns using various charts.
- Generate insights for content strategy and audience engagement.

Tools and Libraries
- **Python**
- **Pandas** – Data manipulation and analysis
- **NumPy** – Numerical computations
- **Matplotlib** – Data visualization
- **Seaborn** – Statistical visualization
- **Plotly Express** – Interactive charts

Dataset Description
The dataset contains social media performance metrics such as:
- Date
- Likes
- Comments
- Shares
- Impressions
- Watch Time
- Engagement Rate
- Followers
- Category
- Post Type
- Country
- Gender
- Age
- Sentiment
- Hashtags

Project Workflow
1. Data Loading
- Imported required Python libraries.
- Loaded the CSV dataset into a Pandas DataFrame.
- Displayed dataset structure, dimensions, and column information.

2. Data Cleaning

- Checked for missing values.
- Filled numerical missing values using median values.
- Filled categorical missing values using mode values.
- Removed duplicate records.
- Converted date columns into datetime format.

3. Exploratory Data Analysis

- Generated summary statistics.
- Examined unique categorical values.
- Created correlation matrix.
- Performed group-wise analysis on:
    - Post Type vs Likes
    - Country vs Impressions

4. Statistical Analysis

Calculated:

- Mean
- Median
- Mode
- Standard Deviation
- Variance
- Quartiles
- Minimum and Maximum value

Visualizations Created

1. Scatter Plot

Likes vs Impressions

- Shows relationship between user engagement and content reach.

2. Line Chart

Daily Engagement Trend

- Displays average engagement score over time.

3. Bar Chart

Posts by Category

- Shows distribution of content categories.

4. Pie Chart

Gender Distribution

- Visualizes audience demographics.

5. Histogram

Age Distribution

- Displays age frequency of users.

6. Box Plot

Engagement Rate Analysis

- Detects outliers and spread of engagement rates.

7. Count Plot

Post Type Count

- Shows frequency of each post type.

8. Bar Plot

Average Likes by Category

- Compares content category performance.

9. Heatmap

Correlation Matrix

- Visualizes relationships among numerical variables.

10. Interactive Scatter Plot

Likes vs Impressions

- Created using Plotly for interactive exploration.

Key Insights

The project identifies:

- Post type with the highest engagement.
- Best-performing content category.
- Country with highest average engagement rate.
- Sentiment trends and audience behavior.
- Correlations between engagement metrics.

Conclusion

This project demonstrates a complete data analysis workflow, including data cleaning, feature engineering, statistical analysis, visualization, and insight generation. The findings can help social media managers and marketing teams improve content strategies and maximize audience engagement.